MD. Mehedi Hasan Maruf

Software Engineer

+8801747090362 | mjmehedihasanoo7@gmail.com | Dhaka (Bangladesh)
mehedi-hasanmaruf | mehedi37 | mehedi-hasan-maruf.me

Education

B.Sc. Engineering in CSE, RUET, Bangladesh (CGPA: 3.75) *Graduated: 08/2026*

Skills

Languages: Python, TypeScript, JavaScript, C#, SQL
CS Fundamentals: Data Structures & Algorithms, Graph Algorithms, Number Theory, Software Design
AI/ML: PyTorch (GNNs), Transformers, LangChain, RAG, NLP, Causal Inference (DoWhy), Bayesian
Systems & Tools: Linux, Docker, CI/CD, Git, Node.js, FastAPI, Django, React, PostgreSQL

Experience

Software Engineer II *Jan 2025 – Present*

Aamira Software Solutions, Kokomo, Indiana, USA (Remote)

- Reduced API response time by 40% by architecting and implementing a TypeScript microservice backend.
- Improved application performance by 60% by profiling and optimizing PostgreSQL/Supabase database queries.
- Increased client response rates by 30% by leading development of a customer relationship management system.

Junior Software Engineer *Nov 2022 – Apr 2024*

Code Studio, Rajshahi, Bangladesh

- Built Python (Django) and JavaScript/TypeScript backends supporting thousands of daily users.
- Maintained 99.8% uptime across 12 Linux servers; cut deployment time by 40% via CI/CD optimization and Docker containerization across 5 client environments.

Research

TRACE: Curriculum Bottleneck Detection & Causal Validation *Jul 2025 – Aug 2026*

B.Sc. Thesis, Dept. of CSE, RUET | *Supervisor:* Farjana Parvin, Assistant Professor

- Designed a graph-mining framework in **Python** fusing prerequisite-graph centrality (NetworkX, **GAT** in PyTorch), Bayesian fail-rate modeling, and aspect-based sentiment analysis (DeBERTa/VADER) into one score.
- Doubled detection quality over the best single-signal baseline (F-measure 0.75, NDCG@10 0.81) on a synthetic benchmark with planted bottlenecks.
- Causally validated results on 23,369 OULAD students with **DoWhy** backdoor adjustment: high engagement raised held-out pass rates by 19.9 pp ($p < 0.01$, E-value 1.89).

Projects

Omnious – Visual Debugging Platform *Nov 2025 – Present*

- Building a debugger with code-graph visualization and AI-assisted debugging (TypeScript, Fastify, tRPC, Next.js).
- Enabled semantic code search with PostgreSQL **pgvector**, ltree, and pg_trgm under Row-Level Security; shipped via Docker and GitHub Actions.

StudentAI – Intelligent Learning Companion *Mar 2025*

- Built a **RAG** learning assistant (**Python**/FastAPI, LangChain, Next.js) with semantic retrieval, conversation memory, multimodal ingestion (PDF, image, video), and quiz generation.

Achievements

- 4th Place, RUET CodeSmash (team, 2023) and CodeSpark (individual, 2022) programming contests; Competitive Programming Mentor, RAPL, RUET (2023), teaching number theory and graph algorithms.